YEAR 7 MATHS · CHAPTER 9 · MIND MAP

Equations

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1 Introduction to equations

An equation says two things are equal. The solution makes it true.

$x + 3 = 7$ is true when $x = 4$

$2x = 10$ is true when $x = 5$

Is $x = 3$ a solution of $2x + 1 = 7$?
$2(3) + 1 = 7$ ✓ yes

2 Solving by inspection

Ask: what number makes this true?

$x + 5 = 12 \Rightarrow x = 7$

$3x = 18 \Rightarrow x = 6$

$\frac{x}{2} = 4 \Rightarrow x = 8$

$x - 4 = 9 \Rightarrow x = 13$

3 Equivalent equations

Do the same to both sides and the equation stays balanced.

$x + 3 = 7$: subtract 3, $x = 4$

$2x = 8$: divide by 2, $x = 4$

$x - 5 = 2$: add 5, $x = 7$

$\frac{x}{3} = 2$: multiply by 3, $x = 6$

4 Solving algebraically

Undo each operation in reverse order, then check by substituting.

$2x + 3 = 11 \Rightarrow 2x = 8 \Rightarrow x = 4$

$5x - 2 = 13 \Rightarrow 5x = 15 \Rightarrow x = 3$

$4x + 1 = 21 \Rightarrow 4x = 20 \Rightarrow x = 5$

Check: $2(4) + 3 = 11$ ✓

5 Equations with fractions

Multiply both sides to clear the fraction.

$\frac{x}{3} = 4 \Rightarrow x = 12$

$\frac{x}{2} + 1 = 5 \Rightarrow \frac{x}{2} = 4 \Rightarrow x = 8$

$\frac{x+2}{3} = 4 \Rightarrow x + 2 = 12 \Rightarrow x = 10$

6 Equations with brackets

Expand the brackets first.

$2(x + 3) = 14 \Rightarrow 2x + 6 = 14 \Rightarrow x = 4$

$3(x - 1) = 12 \Rightarrow 3x - 3 = 12 \Rightarrow x = 5$

$4(x + 2) = 20 \Rightarrow 4x + 8 = 20 \Rightarrow x = 3$

7 Using formulas

Substitute what you know, then solve for what's left.

$A = lw$, $l = 6,\ w = 4$: $A = 24$

$P = 2l + 2w$, $P = 20,\ l = 6$: $20 = 12 + 2w$, so $w = 4$

$D = ST$, $S = 60,\ T = 2$: $D = 120$

8 Solving problems

Use a letter for the unknown, write an equation, solve it, answer in words.

A number plus 9 is 21: $x + 9 = 21$, so the number is 12

3 pens plus a $5 ruler cost $14: $3x + 5 = 14$, so each pen is $\$3$

Rectangle, width w, length $w + 3$, perimeter 26: $4w + 6 = 26$, so $w = 5$: 5 by 8